WORKED SOLUTION

# STEP III 2018, Question 13

Probability and Statistics • Poisson Distribution

**STEP III 2018, Question 13**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q13

## The key idea

Substituting $t = -1$ into a probability generating function attaches a sign $(-1)^r$ to each probability, so averaging $G(1)$ and $G(-1)$ cancels the odd terms and keeps the even ones. The same filtering with the fourth roots of unity $1, -1, \mathrm{i}, -\mathrm{i}$ picks out the multiples of 4. Once each new generating function is in closed form, the mean is just its derivative at $t = 1$, and the question becomes a comparison of elementary functions.

## Solution

Write $p_r = \mathrm{P}(X = r)$, so that $G(t) = \sum_{r \geq 0} p_r t^r$.

**Even values.** Adding the series at $t = 1$ and $t = -1$,

$$G(1) + G(-1) = \sum_{r \geq 0} p_r (1 + (-1)^r) = 2 \sum_{r \text{ even}} p_r,$$

because $1 + (-1)^r$ is 2 for even $r$ and 0 for odd $r$. Dividing by 2 gives the required result. (Both series converge absolutely, since $\sum p_r = 1$.)

**The Poisson generating function.** With $p_r = \mathrm{e}^{-\lambda} \lambda^r / r!$,

$$G(t) = \mathrm{e}^{-\lambda} \sum_{r \geq 0} \frac{(\lambda t)^r}{r!} = \mathrm{e}^{-\lambda} \mathrm{e}^{\lambda t} = \mathrm{e}^{-\lambda(1-t)},$$

using the exponential series, which converges for every $t$.

### Part (i)

The probabilities of $Y$ must sum to 1, so $k$ is the reciprocal of $\mathrm{P}(X \text{ even})$. By the first result,

$$\mathrm{P}(X \text{ even}) = \tfrac{1}{2}(1 + \mathrm{e}^{-2\lambda}) = \mathrm{e}^{-\lambda} \cdot \tfrac{1}{2}(\mathrm{e}^{\lambda} + \mathrm{e}^{-\lambda}) = \mathrm{e}^{-\lambda} \cosh \lambda, \qquad \text{so} \qquad k = \frac{\mathrm{e}^{\lambda}}{\cosh \lambda}.$$

The generating function of $Y$ keeps only the even terms of $G$:

$$G_Y(t) = k \sum_{r \text{ even}} p_r t^r = k \cdot \tfrac{1}{2}(G(t) + G(-t))$$
$$= \frac{\mathrm{e}^{\lambda}}{\cosh \lambda} \cdot \mathrm{e}^{-\lambda} \cdot \tfrac{1}{2}(\mathrm{e}^{\lambda t} + \mathrm{e}^{-\lambda t}) = \frac{\cosh \lambda t}{\cosh \lambda},$$

where $\frac{1}{2}(G(t) + G(-t))$ is the even part of $G$, exactly as in the first result with $t$ in place of 1. This is the required form.

The mean is $G_Y'(1)$:

$$\mathrm{E}(Y) = \frac{\lambda \sinh \lambda}{\cosh \lambda} = \lambda \tanh \lambda.$$

For $\lambda > 0$ we have $\mathrm{e}^{-\lambda} > 0$, so $\sinh \lambda = \frac{1}{2}(\mathrm{e}^{\lambda} - \mathrm{e}^{-\lambda}) < \frac{1}{2}(\mathrm{e}^{\lambda} + \mathrm{e}^{-\lambda}) = \cosh \lambda$, which gives $0 < \tanh \lambda < 1$. Hence $\mathrm{E}(Y) < \lambda$, as required.

#### Part (ii)

Now we need to keep only the terms with $r$ a multiple of 4.

> **Ask yourself** What plays the role of $t = -1$ when we want to filter by multiples of 4 rather than 2?

Use all four fourth roots of unity. For a whole number $r$,

$$1 + (-1)^r + \mathrm{i}^r + (-\mathrm{i})^r = \begin{cases} 4 & \text{if } 4 \mid r, \\ 0 & \text{otherwise.} \end{cases}$$

If $4 \mid r$, every term is 1. If $r$ is odd, the terms cancel in pairs: $(-1)^r = -1$ and $(-\mathrm{i})^r = -\mathrm{i}^r$. If $r \equiv 2 \pmod 4$, the terms are $1 + 1 - 1 - 1 = 0$. So

$$\sum_{4 \mid r} \frac{x^r}{r!} = \tfrac{1}{4}(\mathrm{e}^x + \mathrm{e}^{-x} + \mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x}) = \tfrac{1}{2}(\cosh x + \cos x).$$

(You can also see this directly: $\cosh x = \sum_{r \text{ even}} x^r / r!$, while $\cos x$ has the same terms with signs $+, -, +, \ldots$, so adding them doubles the terms with $4 \mid r$ and cancels the rest.)

Taking $x = \lambda t$ and multiplying by $\mathrm{e}^{-\lambda}$, and then fixing $c$ by requiring $G_Z(1) = 1$, we get

$$G_Z(t) = \frac{\cosh \lambda t + \cos \lambda t}{\cosh \lambda + \cos \lambda}.$$

The denominator is positive for $\lambda > 0$, because $\cosh \lambda > 1 \geq -\cos \lambda$. Differentiating and setting $t = 1$,

$$\mathrm{E}(Z) = G_Z'(1) = \frac{\lambda(\sinh \lambda - \sin \lambda)}{\cosh \lambda + \cos \lambda}.$$

Since $\lambda > 0$ and the denominator is positive,

$$\begin{aligned} \mathrm{E}(Z) < \lambda &\iff \sinh\lambda - \sin\lambda < \cosh\lambda + \cos\lambda \\ &\iff \sin\lambda + \cos\lambda > \sinh\lambda - \cosh\lambda = -\mathrm{e}^{-\lambda} \\ &\iff \sin\lambda + \cos\lambda + \mathrm{e}^{-\lambda} > 0. \end{aligned}$$

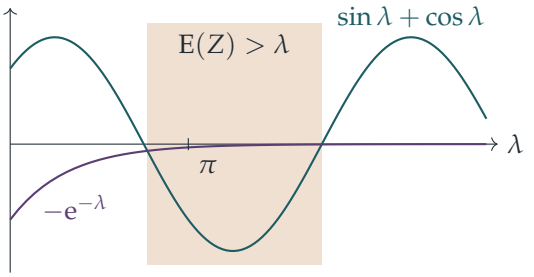
This fails at $\lambda = \pi$: there $\sin\pi + \cos\pi + \mathrm{e}^{-\pi} = -1 + \mathrm{e}^{-\pi} < 0$ because $\mathrm{e}^{-\pi} < 1$. In fact

$$\mathrm{E}(Z)\big|_{\lambda=\pi} = \frac{\pi \sinh\pi}{\cosh\pi - 1} > \pi,$$

since $\sinh\pi > \cosh\pi - 1$ (equivalently $1 > \mathrm{e}^{-\pi}$).

So the answer is **no**: $\mathrm{E}(Z) < \lambda$ does not hold for all positive $\lambda$.

For a fuller picture, write $\sin\lambda + \cos\lambda = \sqrt{2}\sin(\lambda + \frac{\pi}{4})$. The condition fails whenever this curve dips below $-\mathrm{e}^{-\lambda}$. The first such interval runs from about $\lambda \approx 2.42$ to $\lambda \approx 5.49$, and later intervals get closer and closer to $(\frac{3\pi}{4} + 2n\pi,\ \frac{7\pi}{4} + 2n\pi)$ as $\mathrm{e}^{-\lambda} \to 0$.




## Checks

- As $\lambda \to 0$, $\mathrm{E}(Y) = \lambda\tanh\lambda \approx \lambda^2$. This makes sense: $Y$ is almost always 0, and $\mathrm{P}(Y = 2) \approx \lambda^2/2$.
- As $\lambda \to \infty$, $\tanh\lambda \to 1$, so $\mathrm{E}(Y)$ is just below $\lambda$. For large $\lambda$ the Poisson distribution is spread out, so throwing away the odd values hardly moves the mean.
- The computer summed the Poisson series directly and confirmed every generating function and both means. For example, at $\lambda = \pi$ it gives $\mathrm{E}(Y) \approx 3.1299 < \pi$ and $\mathrm{E}(Z) \approx 3.4254 > \pi$. It also located the sign changes of $\sin\lambda + \cos\lambda + \mathrm{e}^{-\lambda}$ at $\lambda \approx 2.419$ and $5.495$.

## Common slips

- Forgetting the normalising constant: $\cosh\lambda t$ alone is not a generating function, because its value at $t = 1$ is not 1.
- Assuming that the answer to (ii) must be "yes" by analogy with (i). Keeping only multiples of 4 can remove more mass below $\lambda$ than above it, for instance when $\lambda = \pi$: the nearest allowed value 4 is above $\pi$ and is much heavier than 0.

- Dividing an inequality by $\cosh\lambda + \cos\lambda$ without checking that it is positive.

## Beyond STEP

Averaging a generating function over roots of unity to pick out the terms in one residue class is called the *roots-of-unity filter* (or *multisection* of a series). It is a basic tool in combinatorics, and it is the simplest case of the discrete Fourier transform. Conditioning a distribution on an event and then recomputing its mean, as here, comes up throughout probability. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions in depth.

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